

Review of JWST observations characterising the earliest galaxies

Abstract

The launch of the James Webb Telescope (JWST) on the 25th of December opens up an exploration of the first galaxies and stars, expected to originate past [$z > 20$, < 180 Myr after the Big Bang], with unprecedented sensitivity and resolution. Initial results have already contributed to tensions with existing models, with observed galaxies being bright and young, with near zero metallicities, and dust. The density of bright, $M_{UV} < -20$, galaxies does not decrease, unlike lower luminosity counterparts, with increasing redshift, hinting at unaccounted for or new physics. The discovery of low redshift interlopers among high-redshift NIRCам galaxy candidates has outlined the importance of NIRSpec spectroscopic confirmation. Tentative spectroscopic detection of a possible Active Galactic Nuclei (AGN) in a [$z = 10.6$, 441 Myr] galaxy might provide constraints on the contested evolution of Super Massive Black Holes (SMBH) and the first generation population III (Pop III) stars that seeded their predecessors. Simulations predict that Pop III stars might contribute significantly to the mass of [$z > 20$, < 180 Myr] blue, $\beta \approx -3$, galaxies, giving JWST the capacity to study these first stars. JWST will build on its predecessor Hubble Space Telescope's (HST), understanding of the reionisation of the Universe between [$5.5 < z < 12$, 1052-372 Myr]. The planned 21cm Square Kilometre Telescope (2029) and gravitational wave Einstein Telescope (2035) will enable a multi-modal study of the early Universe. In this review, the capacities and observations of JWST will be presented, with a discussion of their implications on the evolution of Pop III stars, black holes, and high-redshift galaxies.

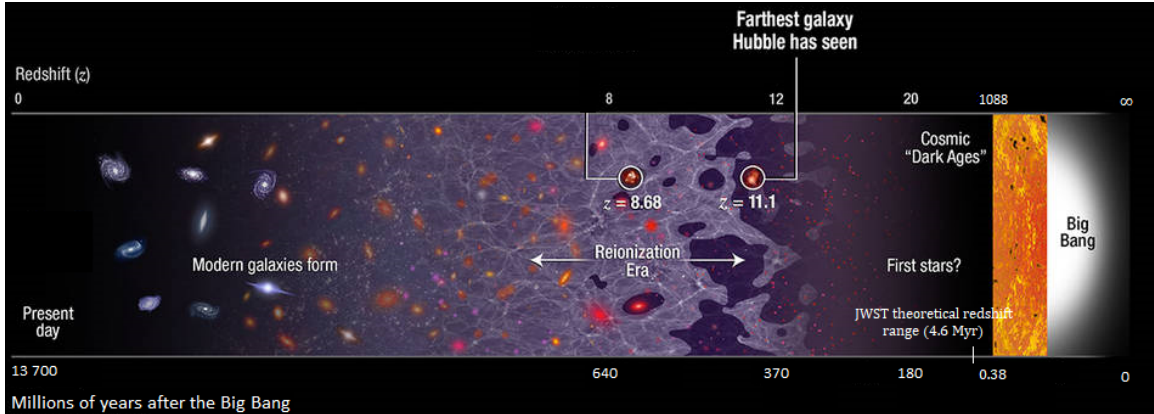


Fig 1: Image from NASA displaying age of the universe in terms of Myr and redshift z

1 Introduction

1.1 Introducing JWST

Before the launch of the James Webb Telescope (JWST), its predecessor the Hubble Space Telescope (HST) could observe galaxies back to [$z \approx 11.1$, ≈ 410 Myr after the Big Bang] at wavelengths 0.1-2.5 μm , as seen in Fig 1 (Oesch et al. 2016). JWST provides the first opportunity to observe the formation of the earliest (high redshift/ z) galaxies, which models expect will host the first generation of stars, Population III stars (Pop III), past [$z > 20$, < 180 Myr after the Big Bang] (Klessen et al. 2023) (Yajima et al. 2022). This is achieved by expanding the wavelength range to 0.6-28 μm , covering near-infrared (NIR) and mid-infrared (MIR), with sensitivities exceeding its predecessors HST and Spitzer.

1.2 Redshifting of observed photons

Photons reaching telescopes from the early Universe are heavily redshifted by the expansion of the Universe, as observed by Hubble (1929). z provides a measurement of how far these photons and spectra are redshifted, $z = (\lambda_{\text{observer}} - \lambda_{\text{rest frame}}) / \lambda_{\text{rest frame}}$. Converting from z to the age of the Universe is out of scope for this review and was calculated using code written by Wright (2006).

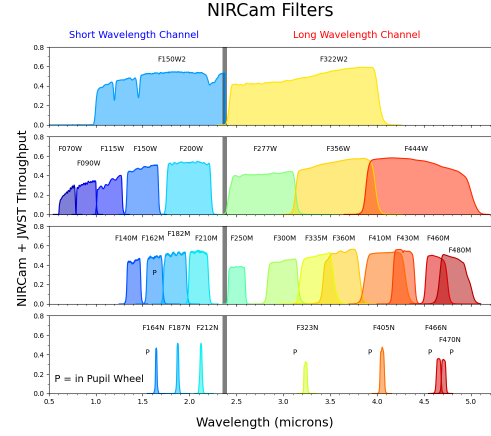
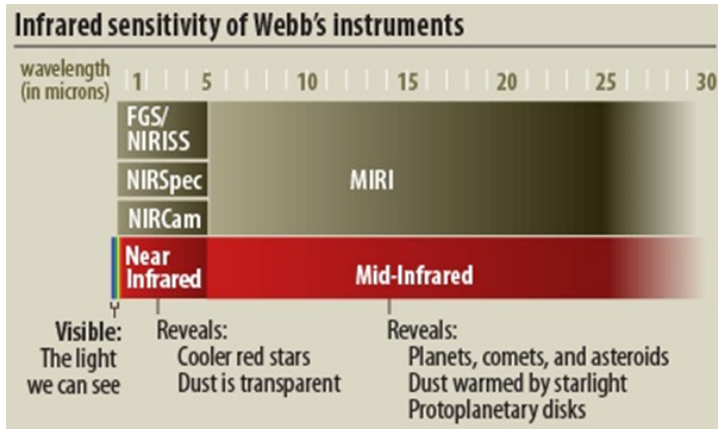


Fig 2: Left:2a displays the wavelength ranges of the four JWST instruments, with MIRI extending into the mid-infrared unlike the other instruments. Right:2b image displays NIRCams filters and their respective wavelengths, with the 2nd row being commonly utilised in high-redshift analysis. Images from NASA.

1.3 How are high-redshift galaxies characterised

Initial redshift estimates for high-redshift galaxies are provided by the Lyman-break, an absence of emission lines below the redshifted 121.56nm (10.2eV) line, caused by the absorption of photons above that energy by neutral hydrogen. The Balmer break provides an indicator of stellar age as only young stars create sufficient UV to saturate the Balmer series absorption. The slope redwards of the Lyman-break, β , acts as an indicator of galaxy metallicity (being any element heavier than He in astronomy) and dust content as these scatter or re-emit photons at lower wavelengths, contributing to reddening. Therefore a low $\beta < -3$ also provides an indicator for the metal and gas free first generation of stars, called Population III (Pop III) stars (Yajima et al. 2022). To directly measure dust content, mid-infrared to millimeter, wavelength observations are required (Hildebrand 1983).

1.4 The Luminosity Function

The luminosity of a galaxy is measured in AB magnitudes, where a decrease by 2.5 represents a 10-fold *increase* in luminosity. This is calculated using, $M_{UV} = -2.5 \log_{10}(f_{\nu}) + 8.90$. Where M_{UV} is the magnitude and f_{ν} is the spectral flux density in units of janskys (Jy) = 10^{-26} WHz $^{-1}$ m $^{-2}$ (Oke et al. 1970). The number density of galaxies at different luminosities, called luminosity function (LF), before JWST, fit the Schechter function (Schechter 1976). The Schechter function, in contrast to the double-power law (DPL) (Yan et al. 2023), assumes significant Active Galactic Nuclei (AGN) feedback, the extremely luminous accretion into a central super-massive black hole (SMBH). The Schechter function and DPL are described by:

$$\text{Schechter} : n(L)dL = \phi(L/L^*)^{\alpha} e^{-(L/L^*)} d(L/L^*)$$

$$\text{DPL} : n(L)dL = \phi(L/L^*)^{\alpha} + (L/L^*)^{\beta} d(L/L^*)$$

Where n is the number density with respect to luminosity (L). L^* is the characteristic luminosity, ϕ is a normalising factor. α determines the faint-end slopes of both functions whilst an exponential cut-off and β describe the bright end for the Schechter and DPL functions, respectively. Meaning that the density of luminous, $M_{UV} < -20$, galaxies drop-off more sharply in the Schechter function due to AGN feedback suppressing the necessary cooling for effective star formation and growth.

1.5 Tensions in Active Galactic Nuclei evolution

The presence of AGNs, as early as [$z=7.5$, 704Myr] (pre-JWST) conflicts with existing SMBH growth models (Bañados et al. 2018). As they evolved from Pop III stars, turning into BHs of masses, $M < 10^2 M_{\odot}$ ($M_{\odot}$ being solar masses), ≈ 700 Myr would be insufficient time to accrete into SMBHs of $M > 10^8 M_{\odot}$ (Larson et al. 2023), at the Eddington rate where $P_{\text{radiation}} / P_{\text{gravity}} < 1$. Alternative models include super-Eddington accretion rates, which are poorly supported due to the lack of cooling mechanisms required to accrete gas at such rates.

1.6 Black Hole formation models

The favoured model is Direct Collapse Black Holes (DCBH), whereby a massive BH seed, $M > 10^5 M_{\odot}$ is formed by the collapse of gas directly into a Pop III star or BH rather than a cluster of smaller stars which will expel gas through radiation and supernovae (SN), called feedback. DCBH requires suppression of the cooling mechanisms that allow Pop III star formation in the absence of metals, which could otherwise radiate energy in poorly absorbed wavelengths. Molecular hydrogen-based cooling mechanisms could be suppressed by Lyman-Werner photons (Prole et al. 2023). This allows the accretion of gas into a Dark Matter minihalo whilst maintaining the thermal pressure preventing the collapse into smaller Pop III stars. Until reaching a critical temperature, where alternative atomic cooling mechanisms take over, and the minihalo collapses directly into a massive Pop III star or BH. These MBH seeds would have the capacity to form the observed AGNs (Agarwal et al. 2012).

Section 2 will present the instruments introduced in JWST, followed by section 3, providing observations made by JWST to date. Section 4 will discuss the timeline of the Universe implied by these observations. Section 5 will conclude the review.

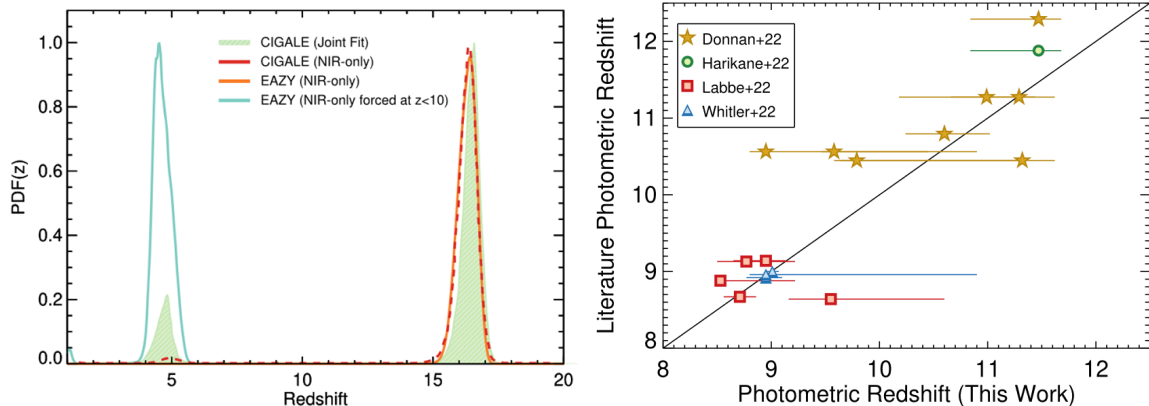


Fig 3: Left: Fig 3a from Zavala et al. (2023) shows probability density from SED fits of CEERS-93316. Red and orange curves display two different SED templates both fitting the galaxy as $z=16$. In green, when sub/millimeter data is incorporated, the fit still prefers $z=16$ but shows significant $z=5$ solution. Right: Fig 3b by Finkelstein et al. (2023) shows their measured vs literature photometric redshifts on candidates with similar measured redshifts. Agreement appears poor at higher redshifts.

Left:3a from Zavala et al. (2023) shows probability density of CEERS-93316 SED fits. Green and red curves show results for with or without NOEMA sub/millimeter data incorporated, with the former indicating a potential $z=5$ fit. Right:3b from Finkelstein et al. (2023) shows the poor agreement between their measured vs literature photometric redshifts.

2 What instruments does JWST introduce

2.1 NIRCam

JWST consists of 4 instruments; Mid-Infrared Instrument (MIRI), Near-Infrared Camera (NIRCam), Spectrography (NIRSpec), and Slitless Spectrograph (NIRISS), their wavelength ranges are shown in Fig 2a (Bagnasco et al. 2007). NIRCam provides camera photometry at 0.6-5 μm . This makes it ideal for detecting potential galaxy candidates, referred to as candidates, and providing redshift estimates using the dropout technique. The Lyman-break causes the galaxy to "drop out" of filters blueward of the redshifted 121.56nm wavelength. For example, the absence of these filters in the second column in Fig 2b would put a lower limit on the estimate redshift; F090W [$z > 7.22$, $< 740\text{Myr}$], F150W [$z > 13$, $< 332\text{Myr}$], F227W [$z > 25.3$, $< 128\text{Myr}$], and F444W [$z > 40$, $< 65\text{Myr}$]. Providing it the potential capacity to observe galaxies containing significant Pop III fractions around [$z \approx 20$, $\approx 180\text{Myr}$](Klessen et al. 2023).

The filters can be used for Spectral Energy Density (SED) fitting whereby the observed luminosities of each filter are collectively fitted to a parameter space of modelled galaxies(Brammer et al. 2008)(Chevallard et al. 2016). The best-fit models provide estimates for a large number of galaxy parameters including; redshift, metallicity, dust, star-formation rate (SFR) history, galaxy age, the probability of each fit, and more. However, due to the limited number and specificity of filters, there is a significant risk of degeneracies. Low redshift galaxies can appear to be high-redshift galaxies (interlopers) by misidentification of the Balmer break as a Lyman break (Zavala et al. 2023). The blue slope and limited emission lines, typical of high-redshift galaxies, can be imitated by nebular hydrogen recombination emission and dust attenuation of weak emission lines, respectively. Furthermore, there are reported issues of selection bias towards bluer galaxies, based on the use of β as a high-redshift selection criterion and altering of luminosities near camera edges(Yan et al. 2023).

2.2 NIRSpec

NIRSpec provides spectroscopy on candidates, with confirmed candidates referred to as galaxies. It has a range of 0.6-5 μm and the capacity to study up to 200 objects at a time. Spectroscopy facilitates concrete determination of redshift, more precise SED fitting, and estimates of parameters determined by emission line ratios or broadening like; metals, dust, AGNs, and outflows.

2.3 MIRI & NIRISS

MIRI extends the JWST range into the mid-infrared, from 5-28 μm . It contains a camera and spectrograph, however, due to lower sensitivities than NIRCam its candidate detection capacity is limited, instead allowing it to measure dust content and further AGN emission lines. NIRISS provides both a camera and slitless spectrograph, imaging at 0.8-2.2 μm , it was designed with exoplanet characterisation in mind.

2.4 Filtering low- z galaxies with spectroscopy

The importance of spectroscopic confirmation of candidates was exemplified by Zavala et al. (2023). NIRCam was used in conjunction with existing Northern Extended Millimetre Array (NOEMA) observations, for dust detection. A candidate, CEERS-93316, from Topping et al. (2022a) was SED fit with only NIRCam, leading to a confident [$z=16$, 250Myr] prediction, as seen in red in Fig 3a. This was confirmed with more conservative selection criteria by Finkelstein et al. (2023). However, when integrating NOEMA into the SED fit, a low but significant probability of a [$z=5$, 1180Myr] fit arose, seen in green in Fig 3a. With the caveat of a predicted 30-40% false positive rate calculated for NOEMA due to the, relatively, low sensitivity. Later analysis with NIRSpec data by Arrabal Haro et al. (2023) determined the galaxy to be $z=5$. Showing how due to the low sensitivity of existing sub/millimeter telescopes, NIRSpec is currently the only reliable instrument for spectroscopic redshift confirmation.

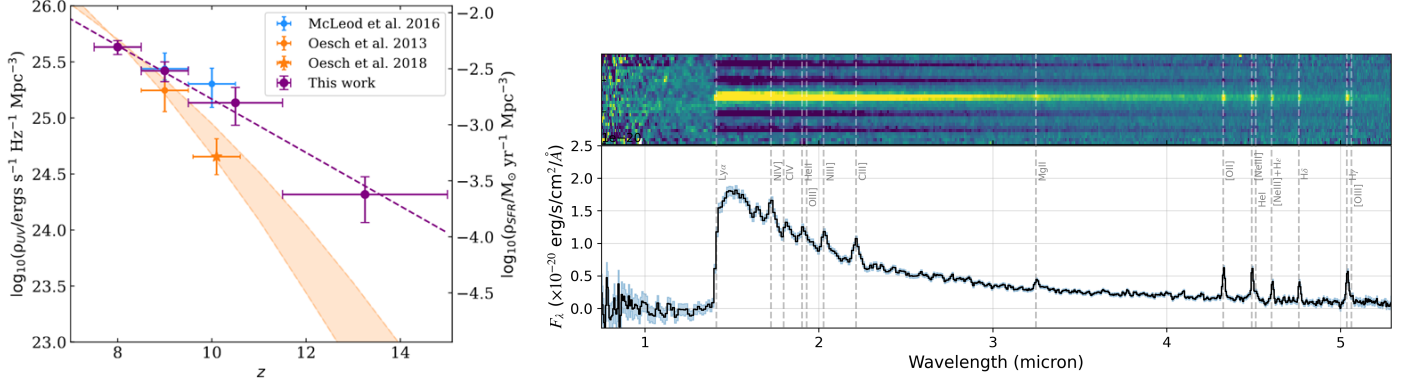


Fig 4: Left:4a shows plot by Topping et al. (2022a) showing lower redshift galaxies fitting to Schechter function in orange, unlike higher redshift Topping et al. (2022a) candidates. Right:4b shows plot by Bunker et al. (2023) providing high S/N spectroscopy, plotting luminosity against wavelength, with rich emission line coverage. Of note are Ly α , N IV, Mg II and He II.

3 JWST observations

3.1 The first spectroscopically confirmed galaxies

JADES ID	GS-z10-0	GS-z11-0	GS-z12-0	GS-z13-0
Full name	JADES-GS+53.15884-27.77349	JADES-GS+53.16476-27.77463	JADES-GS+53.16634-27.82156	JADES-GS+53.14988-27.77650
Exposure time (s)	67225.6	100838.0	67225.6	33612.8
$z_{Ly\alpha}$	$10.38^{+0.07}_{-0.06}$	$11.58^{+0.05}_{-0.05}$	$12.63^{+0.24}_{-0.08}$	$13.20^{+0.04}_{-0.07}$
$z_{H\beta}$	$10.37^{+0.03}_{-0.02}$	$11.48^{+0.03}_{-0.08}$	$12.6^{+0.04}_{-0.05}$	$13.17^{+0.16}_{-0.15}$
EW(C III)] $\text{\AA}$ 2 σ	< 13.8	< 5.9	< 12.4	< 15.2
EW(He II)] $\text{\AA}$ 2 σ	< 14.8	< 6.0	< 13.5	< 15.4
EW(O II)] $\text{\AA}$ 2 σ	< 28.1	< 9.1	< 16.6	< 16.8
2 σ break strength	> 2.04	> 6.85	> 2.48	> 2.79
M_{UV}	-18.61 ± 0.10	-19.34 ± 0.05	-18.23 ± 0.16	$-18.73 \pm 0.06^{\dagger}$
β	-2.49 ± 0.22	-2.18 ± 0.09	-1.84 ± 0.19	$-2.37 \pm 0.12^{\dagger}$
$\log(M/M_{\odot})$	$7.58^{+0.19}_{-0.20}$	$8.67^{+0.08}_{-0.13}$	$7.64^{+0.06}_{-0.39}$	$7.95^{+0.19}_{-0.29}$
$\psi/M_{\odot} \text{ yr}^{-1}$	$1.1^{+0.19}_{-0.16}$	$2.2^{+0.28}_{-0.22}$	$1.8^{+0.54}_{-0.43}$	$1.36^{+0.21}_{-0.23}$
$\log(t/\text{yr})$	$7.54^{+0.25}_{-0.20}$	$8.35^{+0.08}_{-0.17}$	$7.36^{+0.75}_{-0.23}$	$7.84^{+0.23}_{-0.36}$
$\log(Z/Z_{\odot})$	$-1.91^{+0.25}_{-0.20}$	$-1.87^{+0.28}_{-0.19}$	$-1.44^{+0.23}_{-0.22}$	$-1.69^{+0.28}_{-0.31}$
$\bar{\tau}_{*}$	$0.05^{+0.03}_{-0.02}$	$0.18^{+0.06}_{-0.06}$	$0.17^{+0.20}_{-0.09}$	$0.10^{+0.08}_{-0.05}$
ξ_{ion}	$25.46^{+0.07}_{-0.07}$	$25.43^{+0.06}_{-0.06}$	$25.72^{+0.16}_{-0.19}$	$25.47^{+0.10}_{-0.09}$

Table 1: From Curtis-Lake et al. (2022) showing the parameters derived for four galaxies via spectroscopic SED fitting. Of interest are redshift (z), Luminosity (M_{UV}), continuum slope (β), stellar mass ($\log(M/M_{\odot})$), star-formation rate (ψ), oldest stellar age (t), and metallicity (Z).

Six [$z > 10$, $< 478 \text{ Myr}$] candidates have been spectroscopically confirmed to date, Curtis-Lake et al. (2022) provided spectroscopic confirmation of four, with their SED fit parameters presented in Table 1. The following redshifts were measured; [$z = 10.37 \pm 0.03$, $450 \pm 0.02 \text{ Myr}$], [$z = 11.48 \pm 0.08$, $395 \pm 0.06 \text{ Myr}$], [$z = 12.6 \pm 0.05$, $347 \pm 0.02 \text{ Myr}$], and the highest redshift confirmed to date being [$z = 13.17 \pm 0.16$, $326 \pm 0.04 \text{ Myr}$]. These galaxies are less bright compared to the, heavily biased, abundance of bright, $M_{UV} < -20$, observed galaxies and candidates. The metallicities are low, $Z < 0.1$, but not near zero, indicating no significant Pop III presence (Yajima et al. 2022). The measured $\beta < -2$, except for GS-z12-0, supports this.

GS-z12-0 has the lowest luminosity at $M_{UV} = -18.23 \pm 0.16$, which might contribute to the greater uncertainties in its parameters. The stellar masses and SFRs vary significantly, from the most massive, $M = 10^{8.67 \pm 0.13} M_{\odot}$, GS-z11-0, with the highest SFR, $\psi = 158 \pm 58 M_{\odot} \text{ yr}^{-1}$, to the least massive GS-z10-0 at $M = 10^{7.58 \pm 0.20} M_{\odot}$ and SFR, $\psi = 12.58 \pm 5.67 M_{\odot} \text{ yr}^{-1}$. The stellar ages range from 22.9 to 69.0 Myr, being relatively young like many $z < 10$ galaxies, except for GS-z11-0 at $223.8^{+40}_{-72} \text{ Myr}$, which would theoretically indicate that it originated around [$z = 20.5^{+4}_{-4.5}$,

$174^{+72}_{-40} \text{ Myr}$], providing a tentative indication of galaxy formation before $z > 20$. However as galaxy age predictions are reliant on Balmer breaks, which are redshifted out of the highest wavelength filter F444W at [$z > 11$, $< 420 \text{ Myr}$], and which have yet to form in young starburst galaxies, age estimates are unreliable. Curtis-Lake predicts moderate dust attenuation in GS-z11-0 and GS-z12-0, however as the signal-to-noise (S/N) ratio of the spectra was low, no emission lines were detectable. As dust is best predicted using specific emission lines, the dust attenuation is very uncertain.

3.2 Maisies galaxy

Maisies galaxy was spectroscopically confirmed by Arrabal Haro et al. (2023) at redshift [$z = 11.44 \pm 0.09$, $400 \text{ Myr} \pm 0.005$], slightly lower, $\delta_z = -0.36$, than the NIRCcam photometric redshift by Finkelstein et al. (2022). Due to low S/N in the NIRSpec spectra, no clear emission lines were reported other than Lyman α potentially indicating outflows of gas driven by radiation pressure. Maisies galaxy is bright, with $M_{UV} = -20.32 \pm 0.08$, a blue slope of $\beta = -2.47 \pm 0.09$, and a significantly higher mass of $M = 10^{8.5 \pm 0.44} M_{\odot}$ than its Curtis-Lake counterparts. It has a highly uncertain SFR $= 2.1^{+4.8}_{-2}$, age of $18^{+18}_{-9} \text{ Myr}$, compact half-light radius of $R_e = 0.34 \text{ kpc} \pm 0.014$, and a low dust attenuation of 0.07.

3.3 GN-z11

GN-z11 was the highest redshift galaxy with tentative HST spectroscopic confirmation. In contrast to most high-redshift galaxies and candidates observed, it fits the Schechter function as seen in Fig 4a in orange. Providing the first indications of significant AGN feedback at higher redshifts. NIRSpec data analysed by Bunker et al. (2023) determined GN-z11 to be the, to date, brightest spectroscopically confirmed high-redshift galaxy at $M_{UV} = -21.58 \pm 0.04$, β of -2.41 ± 0.07 . It is more massive, $M = 10^{8.73 \pm 0.06} M_{\odot}$, but younger, $18.6 \pm 10 \text{ Myr}$, than its redshift counterpart GS-z11-0, its SFR is $18.78 \pm 0.81 M_{\odot} \text{ yr}^{-1}$. NIRCcam SED fits were reportedly within 1σ of the NIRSpec measurements, indicating strong overlap.

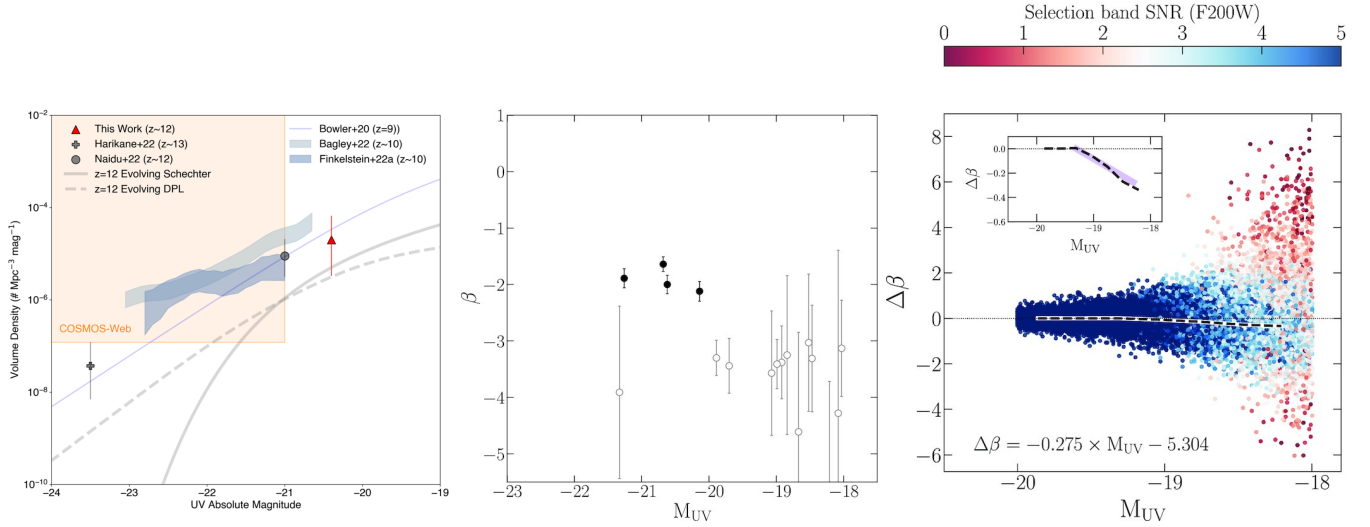


Fig 5: Left:5a from Finkelstein et al. (2022) plots $z=12$ candidates on the LF. Showing data points better fit the $z=9$ DPL function, indicating DPL fit with limited redshift evolution. Middle:5b from Furtak et al. (2023) shows low variance candidates showing higher, $\beta > -2.5$, than high variance ones. Right:5c from Furtak et al. (2023) models JWST observation of simulated galaxies, with $\Delta = \beta_{\text{observed}} - \beta_{\text{intrinsic}}$. Showing that observations at lower luminosity leads to greater scatter/disagreement and a blue bias.

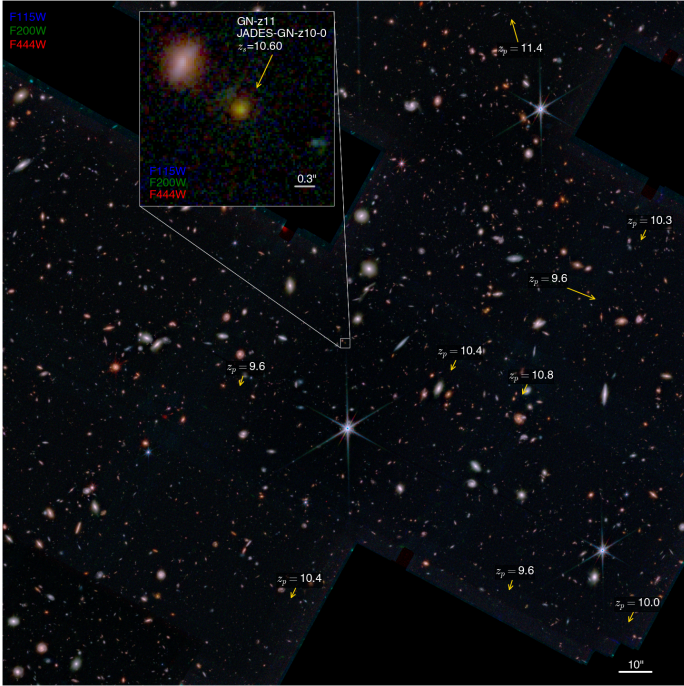


Fig 6: Image from Tacchella et al. (2023) displays NIRC2 image of GN-z11 site. Yellow arrows represent other candidate galaxies in the proto-cluster around $[z > 10, < 478 \text{ Myr}]$.

However, SED fits were performed with the null hypothesis of negligible AGN contribution. This is potentially the highest-redshift AGN candidate to date, with Larson et al. (2023) recently confirming another AGN at $[z=8.6, 568 \text{ Myr}]$ with JWST.

The high S/N emission spectra, seen in Fig 4b, of GN-z11, allow accurate identification of emission lines. Several emission lines are indicative of AGNs like N_{IV}, Ne_{II}, as seen in the $z=8.6$ AGN. However, based on the ratios of C_{III}/He_{II}, C_{III}/C_{IV}, and N_{III}/He_{II}, Bunker et al. concludes that although AGN presence can not be ruled out, the emis-

sion lines indicate a slight preference for stellar dominated solutions. However, our understanding of AGNs is limited at higher redshifts so the presence of an AGN in GN-z11 remains contentious.

The morphology of GN-z11 was determined with NIRC2 by Tacchella et al. (2023), visible in Fig 6. It suggests a highly compact point-spread function component with a half-light radius of $R_e = 0.064 \pm 0.020$ kpc, providing further support for the AGN hypothesis. The spectroscopic presence of Mg_{II} provides indications of outflows or ionised bubbles. The ionised bubble is further supported by the presence of the Lyman α emission line and the overdensity of nearby galaxies, with the potential for ionising overlap, seen in Fig 6. Tacchella et al. (2023) determined the SFR history, concluding that a central point-like source was undergoing a recent intense burst whilst an extended feature surrounding it showed a weak Balmer break. The point-like source was found to contribute to 2/3rd of the rest-UV flux whilst only contributing to 1/5th of the stellar mass, suggesting that the current bright starburst initiated on top of an older stellar population.

3.4 Candidate inferred relationships

Finkelstein et al. (2023) reanalyses candidates presented in the literature and only manages to reproduce the redshifts of $\approx 38\%$ of candidates, these are plotted against their literature redshifts in Fig 3b. Similarly Adams et al. (2023) investigates Atek et al. (2023) and Yan et al. (2023) candidates, reproducing only 23% of redshifts. Implying patterns can only be derived from semi-robust trends, unaltered by the removal of a fraction of candidates.

Many candidates are presented as bright, blue, young, compact, and undergoing starbursts, similar to Maisies galaxy (Furtak et al. 2023)(Naidu et al. 2022)(Yan et al. 2023)(Cullen et al. 2023)(Atek et al. 2023)(Whitler et al. 2023)(Castellano et al. 2022)(Adams et al. 2023)(Topping et al. 2022b)(Topping et al. 2022a)(Finkelstein et al. 2023).

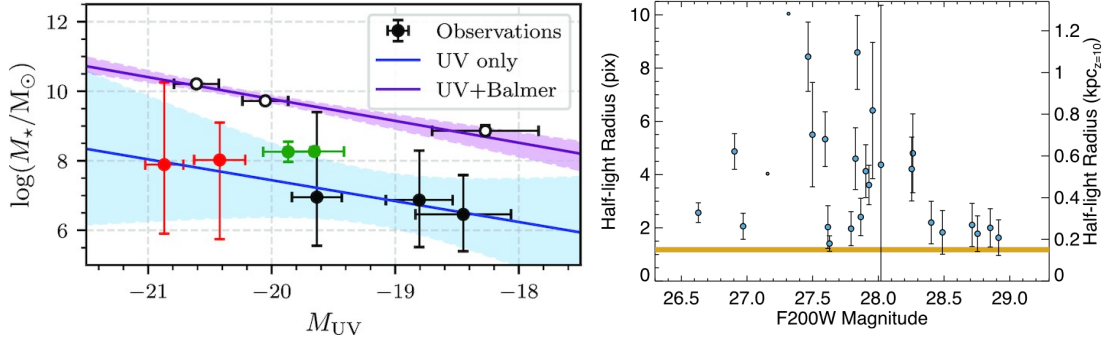


Fig 7: Left:7a from Furtak et al. (2022) shows positive logarithmic relation between mass and luminosity, for both Balmer-break and UV-only candidates, with the latter having lower mass. Right:7b from Finkelstein et al. (2023) shows, if excluding more variable candidates, a trend of half-radius < 0.5 kpc.

With parameters: $M_{UV} < -20$, $\beta < -2$, $t < 100$ Myr, $R_e < 0.5$ kpc, and $SFR > 20$. Fig 5a shows the predicted LF of the selection of $[9 < z < 13, 550-330 \text{ Myr}]$ candidates reproduced by Finkelstein et al. (2023). Unlike at lower redshifts, they show a clear fit to DPL indicating a less aggressive decline in bright galaxy number density and that AGN feedback in most observed high-redshift candidates is insufficient to suppress star formation. Luminosity is linked to stellar mass, as seen in Fig 7a by a logarithmic relationship. The galaxies hosting older stellar populations with Balmer breaks have higher masses at similar luminosities. Fig 7b shows that if data points of high uncertainty are removed, a clear trend showing R_e below or near 0.5 kpc, irrespective of luminosity, emerges. β shows less robust trends. Fig 5b shows that when limiting the analysis to high luminosity candidates with low uncertainties, β does not fall below -2.5 . The $\beta < -2.5$ trend at lower luminosities could be explained by their greater scatter combined with selection bias as seen in the Fig 5c simulation (Furtak et al. 2023). Candidates agree on negligible dust attenuation, explained by the timescale of SN production (10^8 Myr) being greater than the young stellar ages of observed candidates.

4 Discussion on timeline of the early Universe

4.1 Age of galaxies

One of the primary questions when studying high-redshift galaxies is the timeline of their evolution. This would provide constraints on various apparent discrepancies in standard models including Dark Matter, Pop III stars, and SMBHs. The earliest confirmed galaxy is JADES-GS-z13-0, which pushes the observable range of galaxies to at least $[z=13, 330 \text{ Myr}]$, with its oldest stars predicted to form at $[z=17 \pm 230]$. Despite being observed at a lower redshift, JADES-GS-z11-0 is significantly older, moving the boundary up to $[z=20.5^{+4.5}_{-4.5}, 174^{+72}_{-40} \text{ Myr}]$.

4.2 Star bursts and binary stars

The used SED parameters, including ages, do not account for stellar binaries, which due to the low metallicities and high masses are expected to compose a large fraction of high-redshift stars (Sana et al. 2012) (Moe et al. 2019). Binaries have longer lifespans which might alter galaxy age predictions and reduce the LF disagreement. Furthermore, most observed galaxies and candidates appear to show young starbursts, it is unclear whether these are new galaxies or if we're observing a starburst on top of older populations, potentially including Pop III star progenitors which according to Yajima et al. (2022) are expected to make up 44% of stellar mass at $[z=20, 180 \text{ Myr}]$. The consistent detections of these starbursts could be skewing the bright end of the LF whilst also sampling environments with high stellar and SN feedback, contributing to limited or outshone AGN activity. Whether the lack of AGN feedback is temporary or permanent, this could explain the observed fits the DPL.

4.3 Implications of high-redshift AGNs

The observation of AGNs with JWST at $[z=7.5, 704 \text{ Myr}]$ and $[z=8.6, 586 \text{ Myr}]$ provides an opportunity to study both SMBH evolution and Pop III stars indirectly (Bañados et al. 2018) (Larson et al. 2023). If further detections of AGNs, potentially GN-z11, fit to either the super-Eddington or DCHB lines plotted in Fig 8a this could provide constraints on both their evolution and the cooling mechanisms regulating Pop III stars (Klessen et al. 2023).

4.4 Top-heavy Initial Mass Functions

In the DCBH model, cooling and thereby small star formation is suppressed, contributing to a top-heavy initial mass function (IMF), this being the distribution of masses for newly formed stars. A top-heavy IMF reduces LF tension by increasing galaxy brightness and stellar feedback. The high-redshift simulations developed by Yajima et al. (2022) and Behroozi et al. (2015) both succeed in modelling the DPL LF fit, by assuming a top-heavy IMF and a 100% star formation efficiency (SFE), respectively. The support for near 100% SFE from theory and trends is limited due to requiring new cooling mechanisms for Pop III stars.

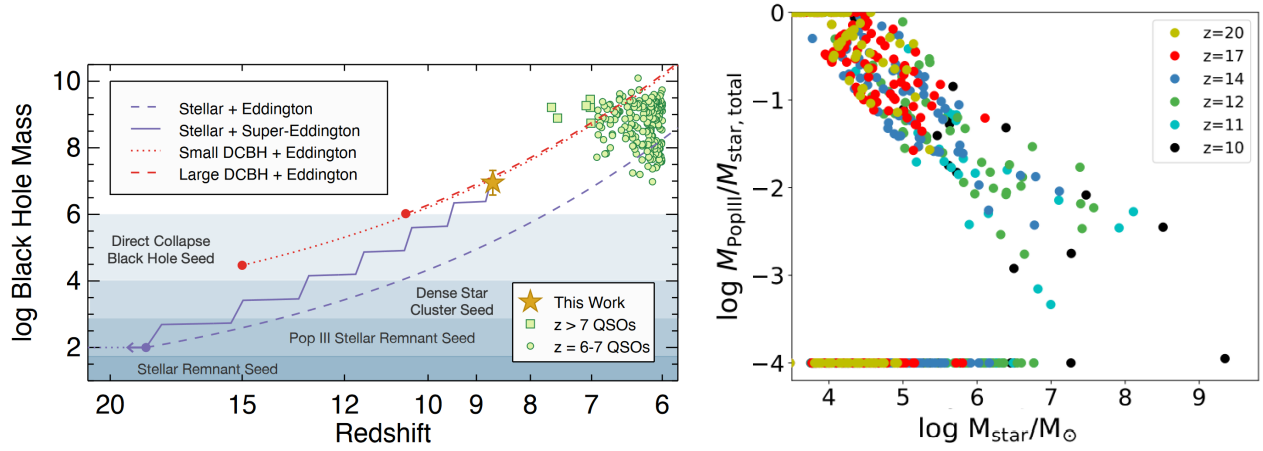


Fig 8: Left:8a from Larson et al. (2023) shows alternative black hole growth models with logarithmic black hole mass against redshift. Only the Super-Eddington (solid blue) and DCBH (dotted red) models are theoretically capable of producing the $M=10^{6.95}M_{\odot}$ [$z=8.6$, 586Myr] black hole. Right:8b from Yajima et al. (2022) shows the mass fraction of population III stars against total stellar mass. Excluding the outliers at the bottom, higher redshift galaxy models like $z=20$ have higher fractions of Pop III stellar mass.

5 Conclusion

JWST has already exposed new tensions and potential resolutions to existing models. The evolution of the LF could be resolved by a variety of models, from selection bias to a different IMF for Pop III stars. However, progress will depend on spectroscopic confirmation to remove low redshift interlopers and observation of emission lines to determine the history of the observed galaxies. NIRCam observations of confirmed galaxies provide insight into their morphology and environment, allowing the future characterisation of large scale galaxy cluster and cosmic web formation based on Dark Matter constraints. Longer NIRSpec exposure times and new high-redshift candidates will expose new extreme galaxies, providing constraints on the LF, high-redshift AGNs, and potentially the first spectroscopy of galaxies populated by Pop III stars. These will be supplemented by other upcoming telescopes like the SKA and Einstein Telescope, providing potential answers to fundamental questions about the nature of dark matter, dark energy, and the transition from a Universe composed of only neutral hydrogen to a chemically and structurally complex one.

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